

Bellcourt Health Administrators: prior authorization under pressure

Data-backed diagnosis and solution. FDE hackathon, US healthcare. Lohith Devaramane, 6 October 2026.

Summary. Bellcourt is late because requests arrive incomplete, and wrong because reviewers cannot find the rule that governs. It is not mainly a staffing problem. I built the **Bellcourt Review Copilot**, an Agentic RAG workspace that runs beside PACE. Code selects the governing rules by client, service and date. A model reads those sections and checks each criterion with quoted evidence. Incomplete requests are caught at intake. A nurse or physician makes every decision, and the decision letter is generated from it. On the 120 cases Bellcourt's own QA team audited, the copilot agrees with the auditor on **97.5%**; the original human decisions agreed on **56.7%**. It made zero wrongful approvals and it cannot deny.

56.7% Decisions correct in QA audit 68 of 120. Contract standard 95%	90.1% Riverbend MA decisions on time Q2 2026. Standard 97%	56.2% Riverbend MA denials overturned 41 of 73. Review trigger 30%	\$1.9M Service credits paid, FY2026 About 21% of forecast profit
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1. The business and the prior authorization process

What Bellcourt is. A third-party administrator (TPA) founded in 2004 in Nashville, about 310 staff, FY2025 revenue \$112.2M. It does not insure anyone. It runs health plans on behalf of others and is paid fees for doing so.

Customer	What Bellcourt does for them	How Bellcourt is paid	What they care about
38 self-funded employers, about 453,000 people. Largest: Harlan Freight Lines	Processes claims, tracks eligibility, answers members, and reviews care requests. The employer pays the medical bills from its own money (governed by ERISA).	\$34 to \$46 per employee per month. About two-thirds of revenue (\$75.1M).	No wasteful approvals (it is their money) and no wrongful denials (their staff complain to HR). Contracts renew every 1 to 3 years.
Riverbend Health Plan: Medicare Advantage (92,400 members, AZ and TX) and Marketplace (61,300 members, GA)	Decides prior authorization requests on Riverbend's behalf (delegated utilization management) since 2023.	Per member per month plus per case. UM fees are \$13.9M, the growth line.	Timeliness (97%), accuracy (95%), appeal outcomes. Riverbend stays accountable to CMS, audits quarterly, and charges service credits.

What prior authorization is. A check before an expensive service such as an MRI or a knee replacement. It answers two separate questions: is the service **covered** under this member's plan, and is it **medically necessary** under written criteria? A nurse can approve. Only a physician can deny on medical necessity. Decisions have legal deadlines counted from receipt: 7 days for Medicare Advantage, 15 days for Marketplace and employer plans, 72 hours if urgent. Bellcourt handles about 310,000 requests a year.

Step	Today	Where it breaks (from the data pack)
1 Intake	Requests arrive by fax (46%), portal (31%), phone (15%), electronic (8%). A coordinator retypes 14 fields from each fax into PACE.	Faxes wait 16.9 hours on average before keying. PACE starts its clock at keying; the contract clock started at receipt.
2 Completeness	Incomplete requests are pended and the provider is contacted.	22.7% arrive incomplete (fax 32.3%). The wait for missing information averages about 3.5 days.
3 Nurse review	Nurse works out which rules apply (plan document, Riverbend addendum, policy version, Medicare rule), reads the record, approves or refers.	14.2 of 38 minutes per case go on finding the rules. PACE criteria screens are months out of date. The library has 1,100 files and file-name search.
4 Physician review	A physician decides proposed denials.	Only 2 of 6 medical directors hold Arizona licenses; Arizona is 66% of Riverbend MA requests.
5 Letter	Staff copy the reason from the PACE note into a Word template.	Letters are vague. Members read a plan exclusion as a medical judgement.
6 Status, appeals	Providers phone for status. A separate team handles appeals.	57% of calls are status checks, 20% abandoned. More than half of appeals are overturned.

The money. Revenue grew 9.3% in three years while clinical staff cost grew 27.7% and intake and call-center cost 23.7%. Operating margin fell from 14.0% to a forecast 8.0%. Riverbend opened a Corrective Action Plan in July 2026, and its contract ends on 31 December 2026.

2. Diagnosis, backed by the data pack

Leadership disagrees about the cause, so I tested each claim against the files: 2,400 requests, 120 QA-audited cases, 90 weeks of staffing, 3,200 calls, 38 knowledge base documents and 8 interviews.

Claim	What the data shows	Verdict
VP Operations: "It's a staffing problem, full stop."	Nurse FTE fell from 57.7 to 50.6 and the nurse stage did slow (32.5 to 38.1 hours). But the on-time rate does not move with staffing (weekly correlation -0.05), and it was already 87.0% in Q1 2025 at 98% of budgeted staff. All 64 late Riverbend MA standard decisions arrived incomplete.	Mostly wrong
VP Operations: overturns come from new information at appeal.	Requests that arrived complete are overturned more often (56.7%) than incomplete ones (48.7%). Denials citing a retired policy version are overturned 91% of the time (30 of 33).	Not supported
Riverbend: "That's a wrong-decision problem."	QA audit: 68 of 120 decisions correct (56.7%) against a 95% standard; 72% on MA cases. 39% of 2026 MA denials cite a retired version.	Supported
Senior nurse: the hard part is which rules apply.	39 of 52 audited errors are wrong-rule errors. Time study: 14.2 of 38.0 minutes locating rules, equal to 18.9 of 50.6 nurse FTE.	Supported
Intake: a third of faxes are missing something.	66 of 69 late decisions arrived incomplete. Requests that arrive complete are on time 99.8% of the time.	Supported

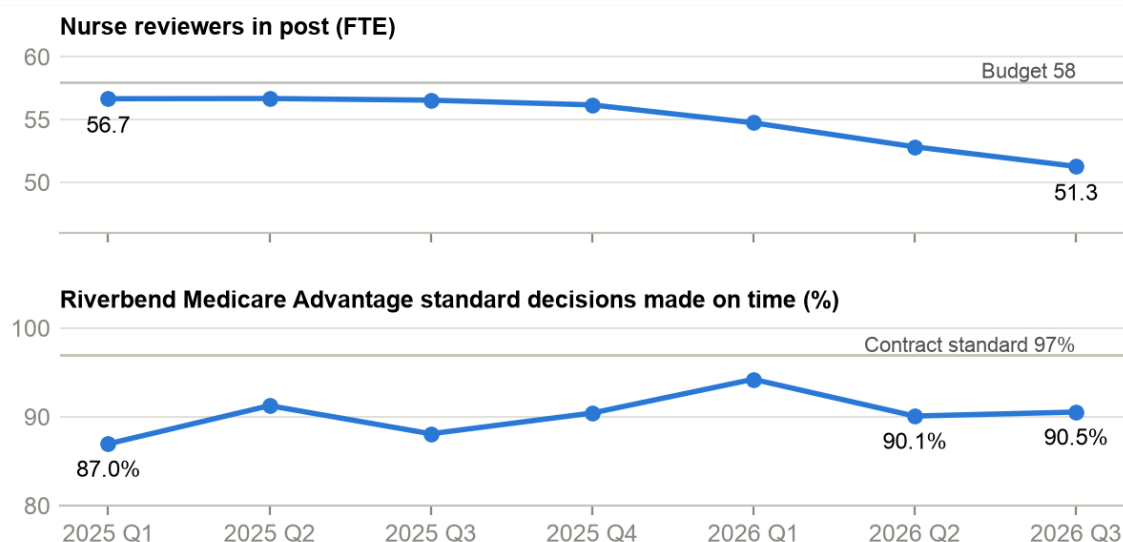


Figure 1. Staffing fell through 2026, but timeliness was already below standard at full staffing and did not fall with it. Sources: nurse_staffing_weekly.csv, pa_requests_2025_2026.csv (662 Riverbend MA standard requests).

Root cause 1: reviewers apply the wrong rule. The rule that governs a case depends on the client, the service and the date of service. None of Bellcourt's tools tracks all three.

- **Retired versions.** 94 of 274 denials in 2026 (34%) cite a policy version that was no longer in force on the date of service.
- **A memo overrode policy.** UM-MEMO-2025-19 told nurses to keep a 6-week rule after the policy changed to 4 weeks. 72 lumbar MRI requests were denied under it; 18 of 21 appeals were overturned. GOV-01 says a memo cannot change criteria.
- **Plan changes did not reach reviewers.** Kestrel began covering bariatric surgery on 1 January 2026. The email went to another team. All 8 requests since were denied as excluded.
- **Medicare rules missed.** 20 glucose-monitor denials for Medicare Advantage members under the stricter Bellcourt rule; all 11 appeals overturned. Without rule errors the MA overturn rate would be 29%, under the 30% trigger.

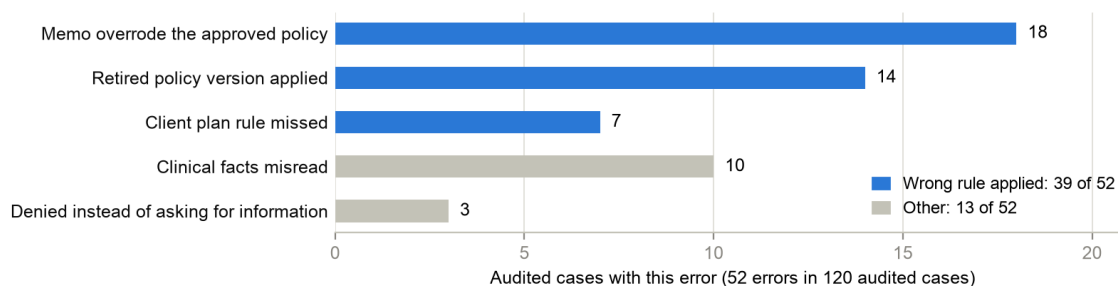


Figure 2. Three quarters of audited errors are failures to find the governing rule, not failures of clinical judgement. Source: qa_audit_sample_2026.csv.

Root cause 2: the clock runs out before a nurse sees the case. Late cases spend 122 of 199 hours waiting for missing information, while nurse review takes about 40 hours whether the case is late or not. Recomputing the real cases: keying faxes within an hour and shortening the wait by a quarter lifts the on-time rate from 90.3% to 97.7%; cutting nurse time by a third reaches only 94.0%.

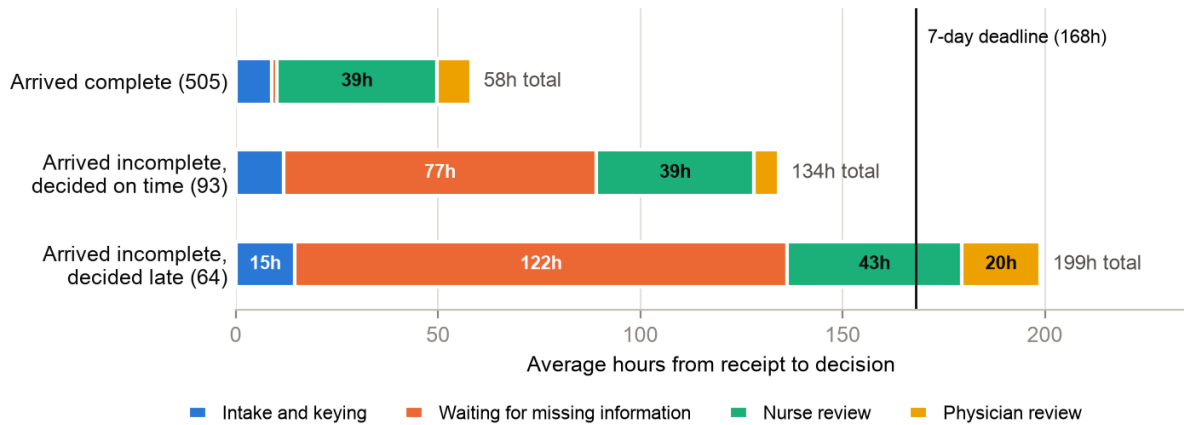


Figure 3. Riverbend MA standard requests: where the hours go. Source: pa_requests_2025_2026.csv.

Caveat stated openly: the pack does not say the QA sample is random, so I do not extrapolate 56.7% to all decisions. The request file shows the retired-version problem independently.

3. Priorities and business impact

Rank	Problem	Evidence	Business impact	Fix and AI pattern
1	Wrong rule applied. Reviewers cannot find which plan document, policy version or Medicare rule governs.	QA accuracy 56.7% vs 95%. MA overturn rate 56.2% vs a 30% trigger. 34% of 2026 denials on a retired version.	Riverbend contract and CMS audit exposure. The "algorithm denies care" headline the CEO fears, in reverse: humans denying wrongly. 18.9 nurse FTE spent hunting rules.	Governing-rule and criteria copilot. Agentic RAG . Built.
2	Clock lost at intake. Fax, incomplete, days of waiting.	66 of 69 late decisions arrived incomplete. MA timeliness 90.1% vs 97%.	Direct cause of the \$1.9M credits (4% of quarterly fees per point) and the Corrective Action Plan.	Read the fax on arrival, check completeness, draft the request for missing items. AI agent step. Built as a slice.
3	No visibility. Providers phone for status; employers get a quarterly PDF.	67% of calls ask where a request is. 20% abandoned.	Call-center cost up 24%. Harlan (\$8.1M a year, satisfaction 5 of 10) renews 1 January 2027.	Status lookup, requirements assistant (plain RAG), dashboard. Next phase.

Why problem 1 first. It is the only fix that makes Bellcourt both faster and more defensible, which is the CEO's brief ("not just faster"). It is the cause Riverbend named, and accuracy below 95% triggers a Corrective Action Plan on its own. It returns nurse capacity without hiring: recovering half the rule-hunting time is worth 9.3 FTE against a vacancy gap of 7.4. Its rule engine is the foundation for problems 2 and 3. Because problem 2 drives the penalties, the build also includes its highest-value slice: catching incompleteness in the first minute.

4. The solution and where it fits

The **Bellcourt Review Copilot** is a signed-in workspace that runs beside PACE and never writes to it. It covers the whole path from request to decision letter.

BELLCOURT SYSTEMS (READ-ONLY). PACE STAYS THE SYSTEM OF RECORD.

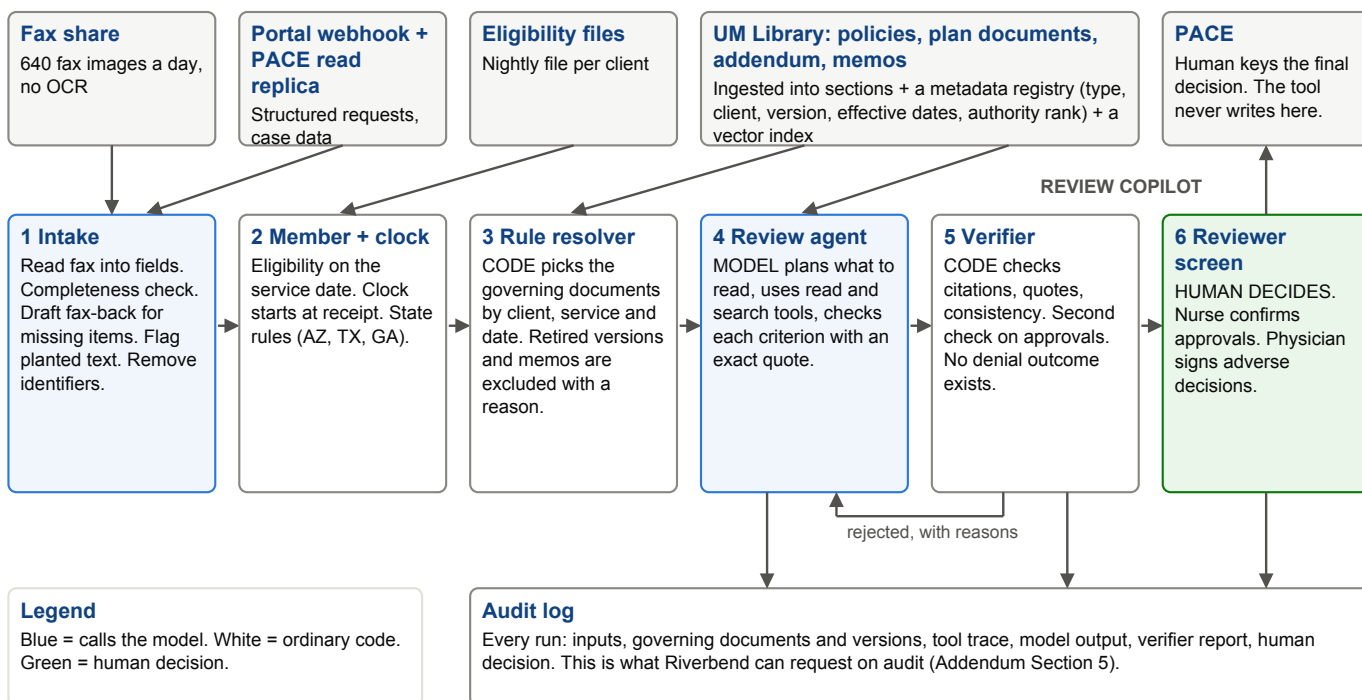


Figure 4. Architecture. Fax share, portal webhook, PACE read replica and eligibility files plug in read-only. Blue boxes call the model, white boxes are code, green is the human decision. Full page: Architecture_Diagram.pdf.

Stage	What happens	Who or what does it
Raise a case	Three ways in: upload a fax image or PDF, upload a case file, or fill the form (phone requests). A fax is read by a vision model into fields; code validates the member ID, service and dates. Every channel becomes the same case record.	Intake, nurse, or the provider feed. Model for faxes only.
Intake check and queue	Missing fields are listed at once and a request to the provider is drafted. The clock starts at receipt. The case joins a priority queue: overdue, urgent, due soon, standard.	Code
Which rules govern	A registry of all 38 documents (type, client, version, effective dates, authority rank per GOV-01) selects the plan document and amendments, the Riverbend addendum and Medicare rule, and the policy version in force on the date of service. Retired versions and memos are excluded and shown with the reason.	Code. The model never chooses a version.
Review	The model plans which sections to read, reads only governing documents, follows cross-references, and marks each criterion met, not met or not documented with an exact quote from the record. Identifiers are removed first.	Model with tools
Verify	Citations must be in force. Quotes must exist in the record. The outcome must match the checklist. Visit-limit arithmetic is recomputed. Every approval gets a skeptical second pass. A failure goes to manual review.	Code plus a second model pass
Human decision	The reviewer sees the recommendation, the reasons, each criterion with its evidence, and the source documents one click away. Approve, pend for information, route to physician, or (physician only) deny. Differing from the recommendation needs a reason.	Nurse or physician
Decision letter and audit	The letter is generated from the human decision: specific reason, provision relied on, appeal rights, Texas AI disclosure. PDF download and email. Every action is logged with user, role and time.	Code, triggered by a person

Why Agentic RAG. Plain RAG retrieves by similarity, but versions 1 and 2 of a policy are near-identical text and the void memo looks highly relevant, so it would repeat today's error. An autonomous agent is not allowed: no tool may deny, delay or modify a request. So code selects the rules, the model reads and reasons within them, code verifies, and a person decides. The four outcomes are approve-ready, need information, route to physician and route as not covered. There is no deny outcome.

5. Evidence it works

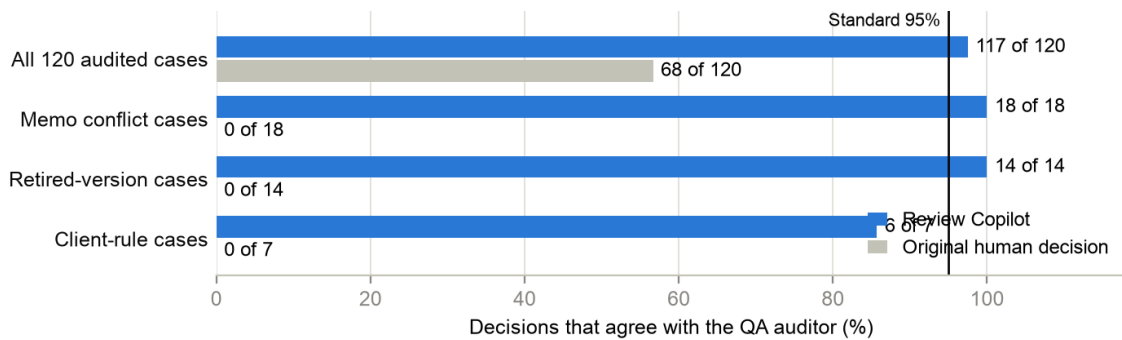


Figure 5. Agreement with the QA auditor's correct decision. The grey bars for the three error groups are zero by definition: those are the cases where the auditor found the human decision wrong for that reason.

Test	Data	Result
Agreement with the QA auditor	120 audited cases with the auditor's correct decision and governing source	117 of 120 (97.5%) . Original human decisions: 68 of 120 (56.7%). Governing document and version correct on 118.
Safety	Same 120 cases	0 wrongful approvals. 0 denials issued (not possible by design).
Open queue	30 live cases including 13 fax images, against my own answer key	29 of 30. Across runs 27 to 29; misses fall on the cautious side.
Adversarial inputs	10 planted instructions (fake approvals, fake memos, "ignore previous instructions"), plus two real faxes in the pack	10 of 10 flagged and ignored. Outcome unchanged.
Wrong and incomplete inputs	Policy time-travel (7), missing therapy record, empty notes, unknown service, no date, unknown member, wrong client	23 of 24 in total. The one failure ended in manual review, not a wrong answer.
Deterministic tests	Resolver, clock, eligibility, de-identification, verifier, sign-in and roles, case workflow, letters	36 of 36 pass, with no model call.

Model: google/gemini-2.5-flash, about 6.5 seconds and a third of a US cent per case. Limits: prompts were tuned while looking at these cases, so a fresh random audit sample is the right next test; only 6 of 38 employer plan documents were in the pack. Details: Evidence_It_Works.pdf.

6. Success metrics

Measure	Today	Target	How measured
Riverbend audit accuracy	72% on MA cases in the QA sample	At least 95%	Monthly QA audit of tool-assisted cases
MA appeal overturn rate	56.2%	Under 30%	Appeals file
MA standard decisions on time	90.1% (Q2 2026)	At least 97%	Clock from receipt, not keying
Denials citing a retired policy version	34% of 2026 denials	Zero	Audit log
Nurse minutes per case	38.0, of which 14.2 finding rules	About 25	Repeat the time study
Hours from fax receipt to missing-information request	About 17 before keying	Under 1	Audit log timestamps
Service credits to Riverbend	\$1.9M a year	Zero	Finance
Adverse decisions made by the tool	n/a	Zero, always	Audit log and tests

7. Compliance and risk

Rule	Source	How the design meets it
No tool may deny, delay or modify a request; a qualified human makes every adverse decision	Riverbend addendum s5; CEO brief	No denial outcome exists. Only the physician role can record a denial, enforced on the server. Requests to providers are drafts a person sends.
60 days' written notice before any AI tool is used	Riverbend addendum s5	Send notice on day 0. Pilot on self-funded plans (ERISA; state AI laws do not bind them) while the notice runs.
Produce inputs and outputs for any case on audit	Riverbend addendum s5	Each review stores inputs, governing documents and versions, tool trace, output and verifier report. Each human action is logged.
Arizona: licensed medical director signs denials. Texas: disclose AI use. Georgia: human review before denial	Addendum s4; REG-02	State flags on the case. Denial letters carry the Arizona sign-off line and the AI disclosure.
Denial notices state the specific reason, the provision relied on and appeal rights	CMS-0057-F; ERISA; SPD s5.5	Letters are generated from the cited provision and the criteria that failed.

Rule	Source	How the design meets it
Patient data is PHI	IT interview; HIPAA	Sign-in and roles. Name, date of birth, phone, record number and member ID are removed before the review step. Production runs in the Azure tenant under the existing agreement.
Provider content is outside Bellcourt's control	Pack README; intake interview	Treated as data. Planted instructions are flagged, cannot be evidence, and do not change the outcome.
Risk	Mitigation	
The model misreads a record	Rules chosen by code; quotes verified; second check on approvals; a person decides; monthly QA on tool-assisted cases.	
The registry goes stale	A policy is not live until its effective dates are in the registry. Plan amendments are filed there by account managers and copied to UM Operations.	
Reviewers over-trust the recommendation	The screen shows the evidence and the excluded documents, not just the answer. Overrides are easy and recorded.	
Riverbend's January 2027 API obligation	Not solved by this build. Named as a risk needing a vendor gateway; the structured case data is a first step.	